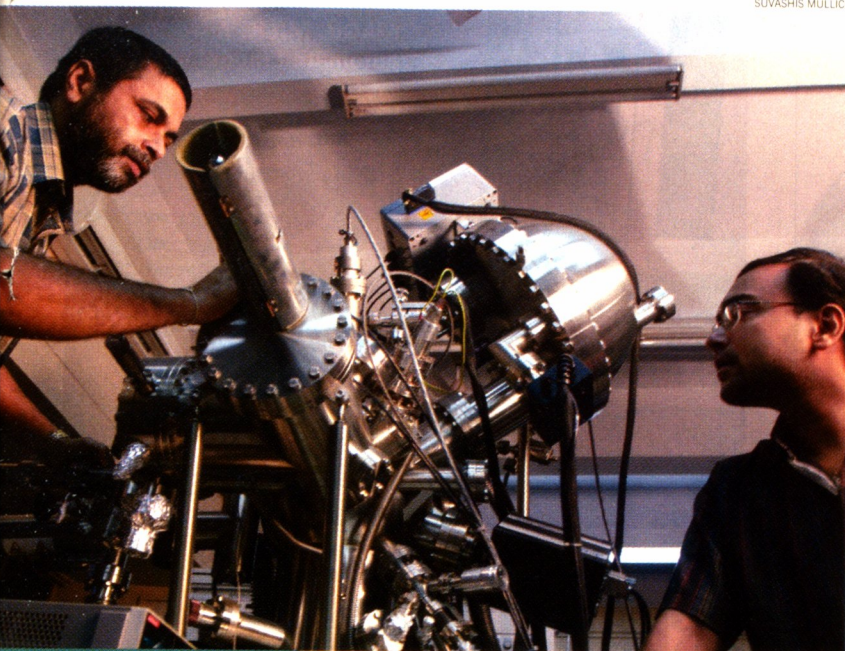




HAWK EYE: Scientists at the nanoscience lab

SAHA INSTITUTE OF NUCLEAR PHYSICS, KOLKATA

ALPHA RESEARCH

Inaugurated by Madam Curie's daughter and Nobel laureate Madame Irene Joliot-Curie in January 1950, the Saha Institute of Nuclear Physics (SINP), Kolkata, conducts research in exotic fields like supernova explosions and black hole entropy. It has also met with success in nanotechnology, biophysics and the gene therapy of chronic myeloid leukemia. With 140 faculty members as well as 150 graduate students and post doctorates, the institute works in varied fields of physics. SINP Director Bikash Sinha says, "Our concept is to concentrate on fundamental research. India underestimates the role of fundamental sciences. Without it, modern technology cannot grow." The physics institute also keeps an eye on global wealth distributions and the dynamics of stock markets through a virgin area called econophysics even as it works on bridging the gap between the traditional education system and specialised research through its centre for advanced research and education. It now plans to install a telescope and set up two new centres—the centre for astroparticle physics and the centre for nanoscience and surface physics—for a more practical learning experience.

by Sudip Bandyopadhyay

AN IDEA IS BORN

In 1949, eminent scientist Meghnad Saha needed a cyclotron for his research. His pursuit to build one led to the setting up of the Institute of Nuclear Physics in the University College of Science, Kolkata, with the active involvement of the then prime minister Jawaharlal Nehru.

INDIAN INSTITUTE OF SCIENCE, BANGALORE

MIND FRAME

It was exactly 100 years ago that Jamshedji Tata founded the institute with the help of the British Government and the Maharaja of Mysore. It was born of an accidental meeting between Tata and Swami Vivekananda on a voyage from Yokohama to Chicago in 1893. Tata was impressed by Vivekananda's views on science and five years later asked him to endorse it. Vivekananda agreed and in 1911, the first batch of students joined the departments of general and applied chemistry, and electrotechnology. Now with 48 departments and over 2,000 post graduate research students, it is difficult to characterise IISc. It is not just a research and applied work institute, nor is it a conventional teaching centre. It is closer to the MIT of the US and made global headlines for its innovation, the Simputer. It has an impressive list of alumni and people like C.V. Raman and Homi Bhabha were closely associated with it. A terror attack in 2005 saw the death of an IIT professor and injuries to four scientists, but its desire to meet new challenges is unimpaired.

by Swagata Sen

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acre or 1.79 sq km is the size of the campus, which is roughly as big as the religious town of Shirdi, in Maharashtra.



PUSHING INNOVATION: P. Balaram, Director IISc